JALAJ SANJAY MEHTA

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PROFESSIONAL SUMMARY

Materials Science Engineer with hands-on experience across electrochemistry, nanomaterials, and semiconductor device characterization, paired with strong Python skills for data analysis, modeling, and tool development. Proven ability to design experiments, build analytical tools, and translate raw data into actionable results.

EDUCATION

B.S. - Materials Science & Engineering | *Columbia University, SEAS*

Minor: Computer Science • Dean's List (All Eligible Semesters) • Bonomi Scholar (2022)

M.S.E. - Materials Science & Engineering | *University of Pennsylvania, SEAS*

RESEARCH & TECHNICAL PROJECT EXPERIENCE

Senior Design Researcher – Battery Technology | *Columbia MSE Dept.*

- Led new product development research on lighter, higher-energy-density electrochemical cells, using qNMR and DSC to build phase diagrams of organic liquid electrolytes
- Conducted comprehensive literature review to define product-relevant performance benchmarks for SPAN anodes; fabricated prototype electrochemical cells from scratch

Research Assistant – DNA Crystal Metallization | *Columbia University & Brookhaven National Lab*

- Established a novel procedure for metalizing DNA crystal lattices; synthesized and characterized crystals via SEM, TEM, FIB, XRD, and EDX -- translating raw data into actionable next steps
- Developed a data processing method to derive mechanical properties from nanoindentation data, producing reusable analytical tools for the research team

Undergraduate Fellow: TEM Image Processing & Analysis | *NETL (Mickey Leland Energy Fellowship)*

- Wrote a Python program for TEM image processing using 3D matrix mathematics to determine crystal grain misorientation; packaged the tool with a Tkinter GUI for broad team use

Researcher: Polymer Engineering | *SUNY Stony Brook - Garcia Research Program*

- Developed flame retardant hydrogel that incorporated RDP and starch into polymer matrix
- Published in ACS Biomacromolecules (2021); presented at the Fall 2019 Materials Research Society Conference, Boston

PROFESSIONAL EXPERIENCE

College Technical Intern – Transistor Process Control Management | *Northrop Grumman Mission Systems*

- Built and validated a virtual-source transistor model in Python against measured test data, delivering a tool adopted by the engineering team
- Developed a user-facing application to analyze and visualize bulk transistor datasets across multiple file formats, accelerating the team's data-review workflow
- Derived key device parameters (I-V curves, capacitance, conductance, power) from raw measurements; incorporated cold-temperature carrier freeze-out constraints into the model

Product Special Projects Lead – Legal AI Systems | *August*

- Analyze user behavior and AI assistant usage patterns to surface bugs, prioritize feature improvements, and inform roadmap decisions

SKILLS & COMPETENCIES

Technical: Python (image processing, data analysis, GUI development), Java, C, SQL, Arduino, MATLAB, Onshape, LaTeX, Microsoft Excel & PowerPoint, ImageJ, Ultimaker Cura, Gwyddion

Characterization: SEM, TEM, FIB, XRD, EDX, NMR, DSC, nanoindentation, lithography, nanofabrication

Product & Strategy: Product roadmap planning, new product introduction (NPI), market research, competitive analysis, cross-functional project management, voice-of-customer, requirements documentation

Languages: English (native), Gujarati (fluent), Spanish (proficient)